

Familia de V.A. Binomiales.

$$vi) \text{Var}[X] = E[\overset{H(x^2)}{X^2}] - E[\overset{H(x)}{X}]^2$$

$$\text{Var}[X] = E[X] - (np)^2 \quad \text{por VI)$$

$$\text{Luego para } E[H(x^2)]$$

$$= \sum X^2 \binom{n}{x} p^x (1-p)^{n-x} \quad x^2 \rightarrow H(x) \text{ y } e^t \rightarrow g(t)$$

$$\text{aplicando } m_x(t) = E[e^{tx}]$$

$$= \sum e^{tx} \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \sum e^{tx} \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \sum \binom{n}{x} (e^t p)^x (q)^{n-x}$$

por coeeficiente binomial

$$= (e^t p + q)^n$$

$$\text{aplicando inversa } E[X^k] = \frac{d^k m_x(t)}{dt^k} \bigg|_{t=0}$$

$$= \frac{d^2 [(e^t p + q)^n]}{dt^2} = \frac{d}{dt} [n p e^t (e^t p + q)^{n-1}]$$

$$= n p [e^t (e^t p + q)^{n-1} + p e^t (n-1) (e^t p + q)^{n-2}] \bigg|_{t=0}$$

$$= n p [(p+q)^{n-1} + p(n-1)(p+q)^{n-2}]$$

$$= n p [1 + p(n-1)]$$

$$= n p + n^2 p^2$$

$$= n p (1-p) + n^2 p^2 = n p q + n^2 p^2$$

$$\Rightarrow \text{Var}[x] = E[x^2] - E[x]^2$$

$$= npq + \cancel{n^2 p^2} - \cancel{n^2 p^2}$$

$$\underline{= npq}$$